TSA 2018

Problem Solving

Questions taken from the TSA Oxford 2018 Section 1 question paper

Questions	Total marks	Time limit
25	25	45 minutes

Instructions

- You have **45 minutes** to answer all **25 questions**.
- Each question is worth **one mark**. There are no penalties for incorrect answers, so attempt every question.
- Each question has five options, A to E. Choose the one option you consider correct.
- Calculators are **not** permitted. You may use this paper for rough working.
- Questions are numbered 1 to 25 for this set. The number in brackets shows the question's number in the original TSA 2018 paper, e.g. **1 (Q3)**.

How the time limit was set

The time limit follows the TARA Problem Solving module (22 questions in 40 minutes), scaled to the 25 questions in this set: 25 × 40 ÷ 22 = 45.45 minutes, rounded to the nearest minute (45 minutes).

Score: _____ / 25

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1 (Q2) There are 24 pupils in a class. They decided to help the local park by planting a total of 24 plants, a mixture of birch trees and roses. Each girl planted three roses, and every three boys planted one birch tree between them.

How many more boys than girls are there in the class?

A 3

B 6

C 8

D 12

E 18

2 (Q6) Jake has a 500 ml bottle of orange squash that he has made according to the instructions on the bottle of concentrate. The instructions dictate that he should add 4 parts water to 1 part concentrate. He accidentally spills his squash, and now there is only 400 ml in the bottle. He then tops up the remainder with concentrate.

What percentage of the squash in his 500 ml bottle is now concentrate?

A 18%

B 20%

C 28%

D 36%

E 40%

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3 (Q7) Baytree Island's Council is concerned to retain a balance of skilled professional workers amongst the population, and monitors migrants in and migrants out on an annual basis. There is currently some concern about the trend, and the Chief Administration Officer was asked to prepare a statistical table for the next Council meeting, as shown below:

	total			*professionals*			*other employed*			*economically inactive*		
in	**P**	**M**	**F**	**P**	**M**	**F**	**P**	**M**	**F**	**P**	**M**	**F**
2017	174	92	82	44	32	12	33	19	14	97	41	56
2016	154	83	71	45	32	13	24	14	10	85	37	48
2015	202	101	101	44	35	9	38	18	20	120	48	72
2014	201	107	94	55	40	15	36	21	15	110	46	64
2013	201	102	99	59	41	18	32	15	17	110	46	64
out	**P**	**M**	**F**	**P**	**M**	**F**	**P**	**M**	**F**	**P**	**M**	**F**
2017	159	94	65	65	49	16	32	22	10	62	23	39
2016	232	132	100	67	50	17	60	38	22	105	44	61
2015	260	136	124	68	49	19	66	37	29	126	50	76
2014	183	89	94	50	32	18	35	19	16	98	38	60
2013	165	80	85	51	32	19	35	16	19	79	32	47

Key: P = persons, M = males, F = females

In which year did the island have the greatest overall loss of professionals?

A 2013

B 2014

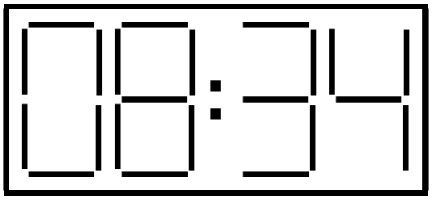
C 2015

D 2016

E 2017

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4 (Q8) Most railway stations have digital clocks that display the time in 24-hour format. I recently undertook a train journey from Exeter to Aberdeen. As the train left Exeter, the clock on the platform showed:



I lost track of time as I read a book and then dozed. I eventually became aware of the train coming to rest and the guard announcing 'Carlisle'.

I opened my eyes, and found that I could only see the top of the platform clock (because of the hoarding in the way) as follows:



Approximately what time was it, to the nearest hour, when the train stopped at Carlisle?

A 1 o'clock

B 2 o'clock

C 3 o'clock

D 4 o'clock

E 5 o'clock

5 (Q12) Below is a table showing the value of Lucy's shares. They started at £3.00 in 2011 and in 2016 they were worth £22.68.

year	*share value*
2011	£3.00
2012	£4.20
2013	£6.30
2014	£9.45
2015	£15.12
2016	£22.68

When did Lucy's shares have the biggest percentage increase?

A 2011–2012

B 2012–2013

C 2013–2014

D 2014–2015

E 2015–2016

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6 (Q13) I am redecorating my house and need to buy wallpaper for one of the rooms. I need to buy 54 m of wallpaper in total and want to pay the lowest price. The table below shows the details of the five types of wallpaper I want to choose between.

wallpaper type	*roll length*	*cost per roll*	*special offers*
woodchip	2 m	£25	every fifth roll is free
vymura	3 m	£35	10% discount for orders over £500
anaglypta	3 m	£40	20% discount for orders over £500
embossed	4 m	£50	25% discount on all orders
fabric effect	5 m	£100	buy one, get one free

Which type of wallpaper should I choose?

A woodchip

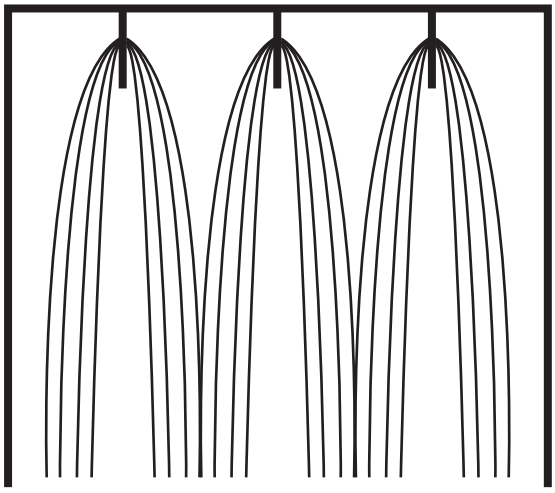
B vymura

C anaglypta

D embossed

E fabric effect

7 (Q14) The diagram below shows the method of construction of a book with four folded sheets in each block, and three blocks in total. Page one is the first page of the first block of four sheets. At the centre of each block, you can see the stitching that binds the book together.



A book is to be made using the same technique but with eight sheets of paper in each of the three blocks.

Which of the following pairs of pages could be in the centre of a block (that is, if opened there, the stitching would be visible)?

A 46 and 47

B 47 and 48

C 48 and 49

D 63 and 64

E 64 and 65

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8 (Q18) Five girls are collecting shells on a beach. They are putting them in buckets that each hold 24 shells. At the end of the collecting time, they are going to ensure that they all have the same number of shells. Four of them are only two short of filling their bucket, but the youngest one is lagging behind with her total. Each of the others give her three shells from their buckets in order to equalise the numbers collected.

How many shells did the youngest girl collect for herself?

A 4

B 7

C 10

D 12

E 19

9 (Q19) A Youth Centre runs a number of football teams for boys of different ages. Each season, one prize is awarded to the player who has given the best all-round performance. Five players have been nominated as possible recipients of this year's prize. The manager has decided not to reward any player who has missed two consecutive training sessions more than twice, and the manager will also exclude anyone who has failed to score a goal from a penalty kick more than twice. After these criteria have been applied, the prize will go to the player who has scored the highest number of goals.

player	*no. of years in a team*	*no. of times late for training*	*no. of times missed two consecutive training sessions*	*no. of penalty kicks taken*	*no. of penalties scoring a goal*	*total no. of goals scored*
David	2	3	2	16	13	32
John	3	2	1	10	9	26
Colin	2	4	1	13	9	30
Mike	1	2	2	12	10	29
Graham	2	0	3	8	7	36

Which player will receive the best all-round performance prize?

A David

B John

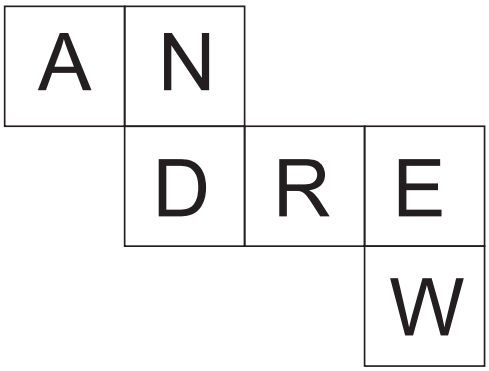
C Colin

D Mike

E Graham

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10 (Q20) The diagram below shows the net of a cube.



The following diagrams show five suggested views of the cube once it has been assembled.



1 2 3 4 5

Which two of the above are possible views of the cube?

A 1 and 2

B 2 and 3

C 3 and 4

D 1 and 4

E 4 and 5

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11 (Q24) I have a digital clock that works in 24-hour format (that is, after 23:59, it goes to 00:00). The patterns for each number are made up of segments, as shown below:



What is the maximum number of segments that can be displayed at once, while still showing a valid time?

A 20

B 23

C 24

D 26

E 28

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12 (Q25) The President of Bugaria is elected using the supplementary vote system, which is conducted as follows:

Voters vote for their first-choice and second-choice candidates (though a second choice is optional).

The two candidates with the most first-choice votes go through to a second stage, whilst the others are eliminated. The second-choice votes for these two from the ballot papers of the eliminated candidates are added to their first-choice votes, and the candidate with the greater total is elected President.

The table below shows a summary of the votes cast for the five candidates in last week's Bugarian Presidential Election. For example, 37 862 voters cast their first vote for Sam and their second vote for Val.

	first-choice votes				
second-choice votes	**Sam**	**Val**	**Kim**	**Pat**	**Jo**
Sam	–	37 304	34 493	38 206	22 871
Val	37 862	–	25 517	27 818	20 143
Kim	23 148	31 690	–	30 247	42 620
Pat	26 493	24 485	29 681	–	31 354
Jo	28 506	27 530	25 308	24 763	–
no second choice	822	1027	1263	952	848

Who has been elected President of Bugaria?

A Sam

B Val

C Kim

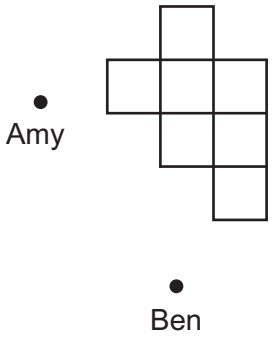
D Pat

E Jo

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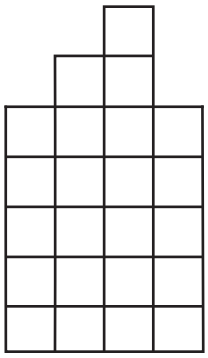
13 (Q26) A number of identical boxes are piled up, waiting to be loaded into a delivery van. Amy and Ben are about to load them into the van.

The appearance of the pile of boxes and the positions of Amy and Ben at present are shown below:



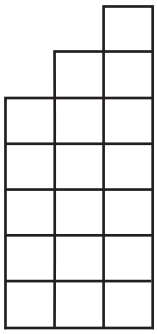
[view from above]

Amy's view of the pile is:

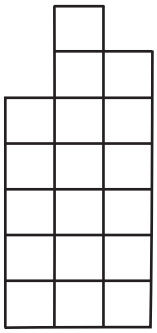


Which one of the following could **not** be Ben's view of the pile?

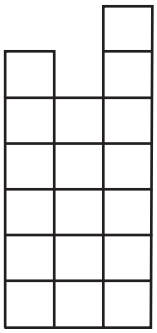
A



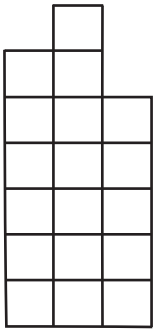
B



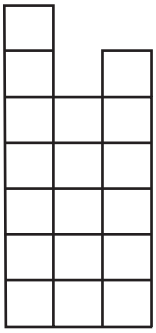
C



D



E



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14 (Q30) Recently, packets of Amblers crisps have had 'money-off' coupons inside them. Some coupons are worth 9p, some are worth 14p and some are worth 20p.

George has been collecting these coupons. He has more 14p coupons than 9p coupons, and more 9p coupons than 20p coupons. The total value of all of his coupons is exactly £1.50.

How many coupons has George collected?

A 9

B 10

C 11

D 12

E 13

15 (Q31) Joanne is studying the period 1900–1950 and has decided to buy a new book to help with her studies. There are five different books available at her local bookshop, the details of which are summarised below:

title	*period covered*	*hardback?*	*illustrated?*	*price*
History of the 20th Century	1900–2000	yes	no	£45
The Illustrated Guide to History	1750–2000	no	yes	£50
Beginner's Guide to History	1700–1850	no	no	£40
History for All	1750–1950	yes	yes	£60
All About History	1800–1900	yes	no	£55

On Tuesday, Joanne chose to buy the cheapest book that would satisfy her requirements, and went away to get the exact amount of money needed to buy it. When she returned the following day to buy her chosen book she found that there was a sale and all of the books had been reduced by 20%. She then decided instead to buy the most expensive book that she could with the money she had, and which still satisfied her requirements.

How much money did Joanne have left over once she had bought the book?

A £0

B £1

C £2

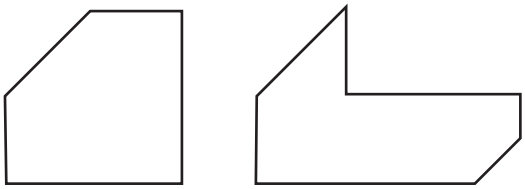
D £3

E £5

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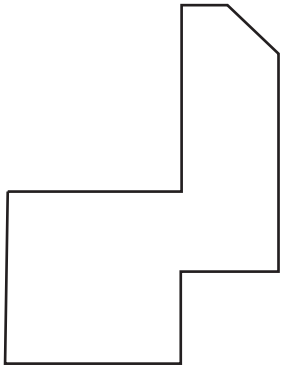
16 (Q32) Connem Ltd. has recently adopted a new company logo.

A large plastic version of the logo has just been delivered to the office, to be attached to the outside of the building. Unfortunately, in transit, the plastic has broken into two pieces, as shown below:

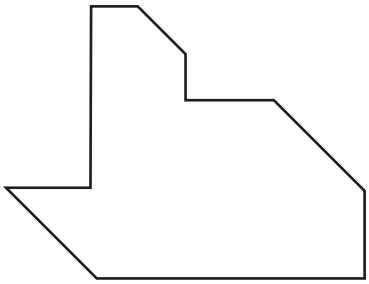


Which one of the following could **not** be Connem's new logo?

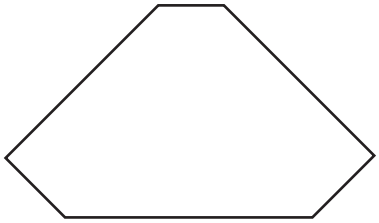
A



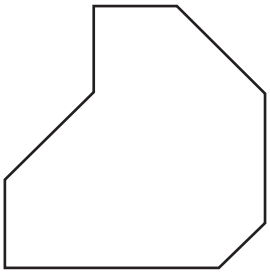
B



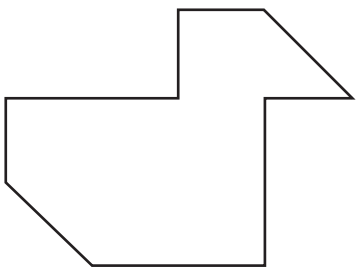
C



D



E



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17 (Q36) At the start of the school year, a village shop stocks and sells a range of stationery items. The table below shows details of the wholesale costs and retail prices for a number of the items sold.

items	*wholesale cost per pack of 12*	*retail selling price per 1 item*	*number sold in one week*
crayons	48p	10p	180
felt tips	72p	12p	150
pencils	24p	6p	200
pens	36p	15p	150
rulers	60p	30p	40

Of the five items sold, which two together made the most profit for the shop owner?

A crayons and pencils

B crayons and pens

C felt tips and pencils

D felt tips and rulers

E pens and rulers

18 (Q37) There is an automatic photograph booth in my local shopping centre. It takes exactly 2 minutes from the time that the money is inserted for the photographs to be taken, then the developed photographs appear 4 minutes later.

Money may only be inserted when the green light is on. This light goes out immediately after money is inserted, and comes back on again exactly 3 minutes later.

This morning, I arrived at the booth just as it was being switched on, and I was sixth in the queue. The first person inserted their money immediately, and the rest of us all inserted our money as soon as we were allowed to.

How long after I arrived at the booth did I insert my money?

A 15 minutes

B 18 minutes

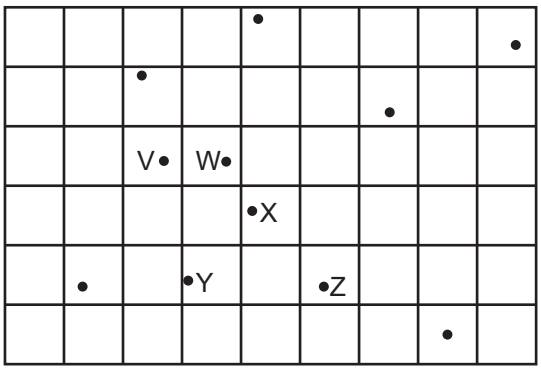
C 21 minutes

D 25 minutes

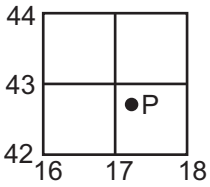
E 30 minutes

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19 (Q38) The diagram below shows part of a map of Ruralia.



A group of students are on an expedition and must meet their teacher at Longwood, grid reference 787634. Unfortunately, they have been given a version of the map with the numbers missing from the gridlines, and the place names have been omitted too. They have a list of grid references for some places on the map, and are trying to work out which point to head for in order to be sure of meeting their teacher.



They have a set of instructions showing how the grid references work:

The grid reference of point P is 172427. Go to vertical line to the west of the point, read the number 17. Estimate how many tenths across the square the point appears at, hence 172. Go to horizontal line to the south of point P, read the number 42, estimate how many tenths up the square the point is, hence 427. This gives 172427 as the complete grid reference number.

Here is the list of the grid references / place names the students have been given.

Hilltop	763613	Riverside	781614
Sunnybank	804613	Topfield	777634
High Moor	792626	Longwood	787634
Highscar	773649	Stakefell	793658
West Pasture	815642	Lower Green	837654
Little Tarn	825605		

Using this information to help you, which of the places marked on the map is Longwood?

A V

B W

C X

D Y

E Z

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20 (Q42) In the main draw of a lottery, six of the balls (which are numbered from 1 to 49) are selected at random, then the numbers chosen are rearranged and displayed in ascending numerical order.

For instance: 3 17 20 29 34 45

In one draw recently, I noticed that the six numbers were made up of a total of ten digits, all different. The lowest number on this occasion was 1 and the highest number was 49, so the range of the six numbers was 48, the greatest possible range for the lottery main draw.

What is the smallest possible range when the six numbers in the main draw of the lottery have a total of ten digits, all different?

A 21

B 23

C 27

D 31

E 32

21 (Q43) Car parking charges are shown in the table below. Brian wants to park his car for a period of 7.5 hours, whilst he is at work. He buys a ticket in the morning when he arrives at the car park. Due to the close proximity of the car park to his place of work, he is able to return to his car to buy a new ticket as many times as needed.

up to	*car parking charges*
1 hour	£0.80
2 hours	£1.20
3 hours	£1.70
4 hours	£2.40
5 hours	£3.50
6 hours	£4.20
7 hours	£5.20
8 hours	£6.20
over 8 hours	£8.20

What is the most Brian can save by returning to the car to buy new tickets compared to buying one 8-hour ticket?

A £1.10

B £1.40

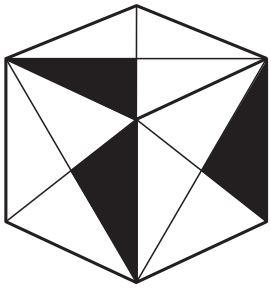
C £1.60

D £2.10

E £3.60

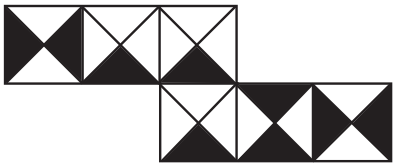
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22 (Q44) Here is a diagram of a cube.

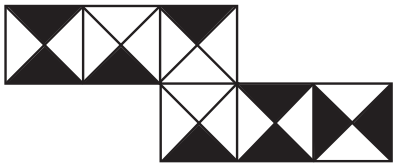


Which one of the following nets **could** be folded to make this cube?

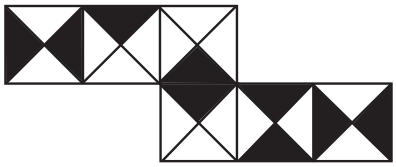
1



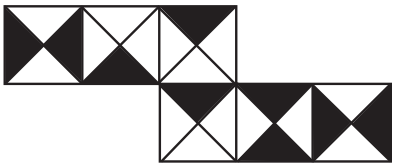
2



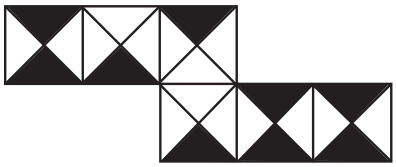
3



4



5



A 1

B 2

C 3

D 4

E 5

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23 (Q48) A car journey begins with a 15 km drive at an average speed of 60 km / hour, then 5 hours on the motorway, including two 15-minute rest stops, travelling at an average driving speed of 80 km / hour, and finally a 20 km drive at an average speed of 40 km / hour from the motorway to the destination.

If the destination is reached at 12:00 midday, at what time did the journey begin?

A 5:18 am

B 5:25 am

C 5:45 am

D 6:15 am

E 6:45 am

24 (Q49) Fiona is going to put some shelves up in her living room. The shelves must each be 1.80 m long and have a depth of at least 40 cm. There will be five shelves in total. The different prices for wood are shown in the table below.

	depth		
length	*30 cm*	*45 cm*	*60 cm*
0.50 m	$0.90	$1.35	$1.50
1.00 m	$1.70	$2.55	$2.80
1.50 m	$2.50	$3.75	$4.20
2.00 m	$3.30	$4.95	$5.50
3.00 m	$4.80	$7.20	$8.00
4.00 m	$6.20	$9.30	$10.00

Fiona can cut pieces of wood to make more than one piece of the correct size, but she wants each of the shelves to be made of one complete piece of wood.

What is the cheapest total cost for which Fiona can make her shelves?

A $21.15

B $23.25

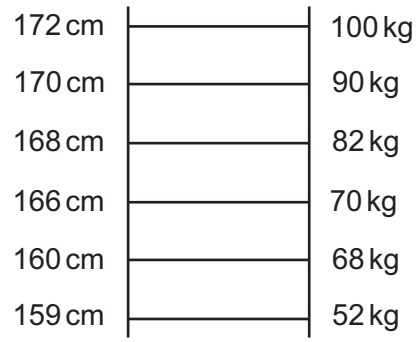
C $23.55

D $24.75

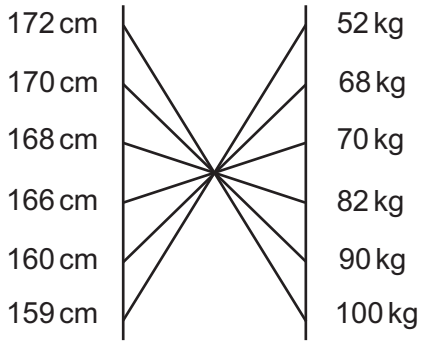
E $25.50

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25 (Q50) Keith likes to draw 'bump charts' to look at relationships. For instance, if 6 individuals were ordered by height, and it happened that the order was the same by weight, the bump chart would look like this:

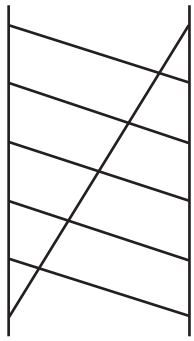


Or if height was maintained for order on the left hand but lightness rather than heaviness on the right hand, the bump chart would be this:

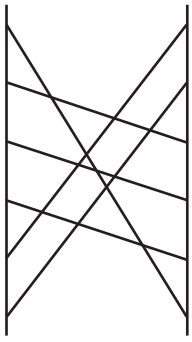


If one axis is reversed in order (as in heaviness goes to lightness), which one of the following bump charts would **not** be symmetrical about a vertical line midway between the axes?

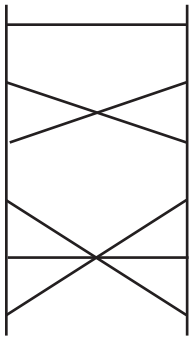
A



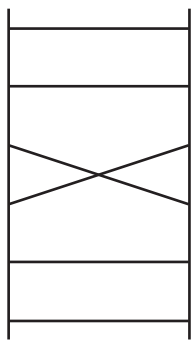
B



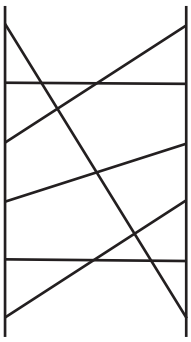
C



D



E



END OF TEST

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